



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-4)

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Computer Networking

Data Communication

Wireless Technologies



Wireless Communication Technology

1. Bluetooth

- Bluetooth is a short-range wireless communication technology.
- It is used in many devices like headphones, keyboard, mouse, printer etc.
- Bluetooth technology uses radio waves between 2.4 GHz to 2.8 GHz.
- It was invented by Dr. Jaap Haartsen (जाप हार्टसेन) in 1994.
- 1st commercial product (wireless headset) in 1999.
- 1st Bluetooth version : version 1.0

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- Current version in 2026 : version 6.3 (relase 6 may 2026).
- 6.3 is updated version of version 6 (version 6 release in sep 2024)
- Range : 10 meter
- Up to 7 devices can be connected via Bluetooth network.
- It is used in Personal Area Network (PAN) or WPAN (wireless PAN).



Bluetooth

2. Wi-Fi

- Wi-Fi stands for Wireless Fidelity.
- It is a wireless technology that enables devices to connect to the internet and communicate with each other in a **Local Area Network** (LAN) or WLAN.
- It was created by John O'Sullivan in 1997.
- It typically ranges up to 10-50 meters indoors and about 100-300 meters outdoors.
- Wi-Fi technology uses **radio waves** between 2.4 GHZ (slow but long distance) , 5 GHz (speed but short distance) and 6 GHz.
- Number of device : depends on device (approx. minimun 10 to 200).



3. Li-Fi

- Li-Fi stands for Light Fidelity.
- It is a technology that uses **light waves** to transmit data wirelessly.
- It was invented by Harald Hass in 2011.
- Li-Fi offers very high-speed data transfer, faster than Wi-Fi.

- It uses **LED** (light signals) for sending data and **photo diode** for receiving data.
- It is useful in places such as hospitals and aircraft cabins.



❖ How LiFiMAX works

wireless communication, LiFiMAX, comprises of 4 key products:



LiFiMAX® Access Point **1.**

The LiFiMAX® Access Point enables a secure connection for user devices. The data received by a router is transmitted to the LiFi access point, which in turn transmits it to nearby antennas as light signals.



Photonic Antenna **2.**

The photonic antenna is positioned to broadcast data-carrying LiFi signals in the internet coverage area. Each antenna supports up to 16 user devices per Access Point, ensuring expansive coverage



Key/Dongle/Receiver **3.**

The Key/Dongle/Receiver is a LiFiMAX® receiver that picks up light signals in the environment and converts it into electrical information that can be interpreted by computers.



Tablet **4.**

The tablet is another LiFiMAX® receiver redefining the user experience for safe and effective communication with the Photonic Antenna.





4. Wi-Max

- Wi-Max stands for Worldwide Interoperability for Microwave Access.
- Founded in 2001.
- It is a wireless communication technology for high-speed internet over long distances.
- Wi-Max can cover areas up to 30 miles/ approx. 50km, much more than Wi-Fi.
- Frequency commonly operates around 2–3.5 GHz.
- Wi-Max use in Wireless Metropolitan Area Network (WMAN).

CPE/antenna



CPE = Customer premises equipment

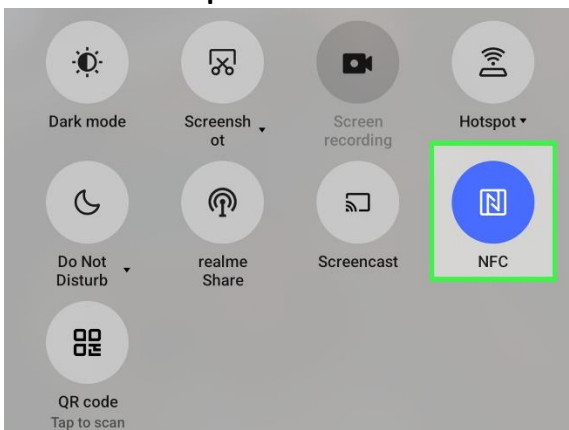
5. HotSpot / wifi hotspot

- A hotspot is a location or device that provides internet access through Wi-Fi.
- It is created using a router or a mobile device with internet-sharing capability.
- They allow multiple devices to connect to the internet wirelessly.
- It is also called Wireless Access Point Technology or wifi hotspot.
- Range usually around 10–30 meters, depending on the device and environment.



6. NFC

- NFC stands for Near field communication.
- It is short range wireless communication technology.
- It generally works within a few centimeters.
- It is commonly used for contactless payments and access cards.
- Smartphones can use NFC to exchange small amount of data.



7. Zigbee

- Zigbee is a low-power wireless communication technology.
- It is mainly designed for IoT and smart-home devices.
- It consumes very little power.
- It is used to smart light, sensors, switches and home automation.

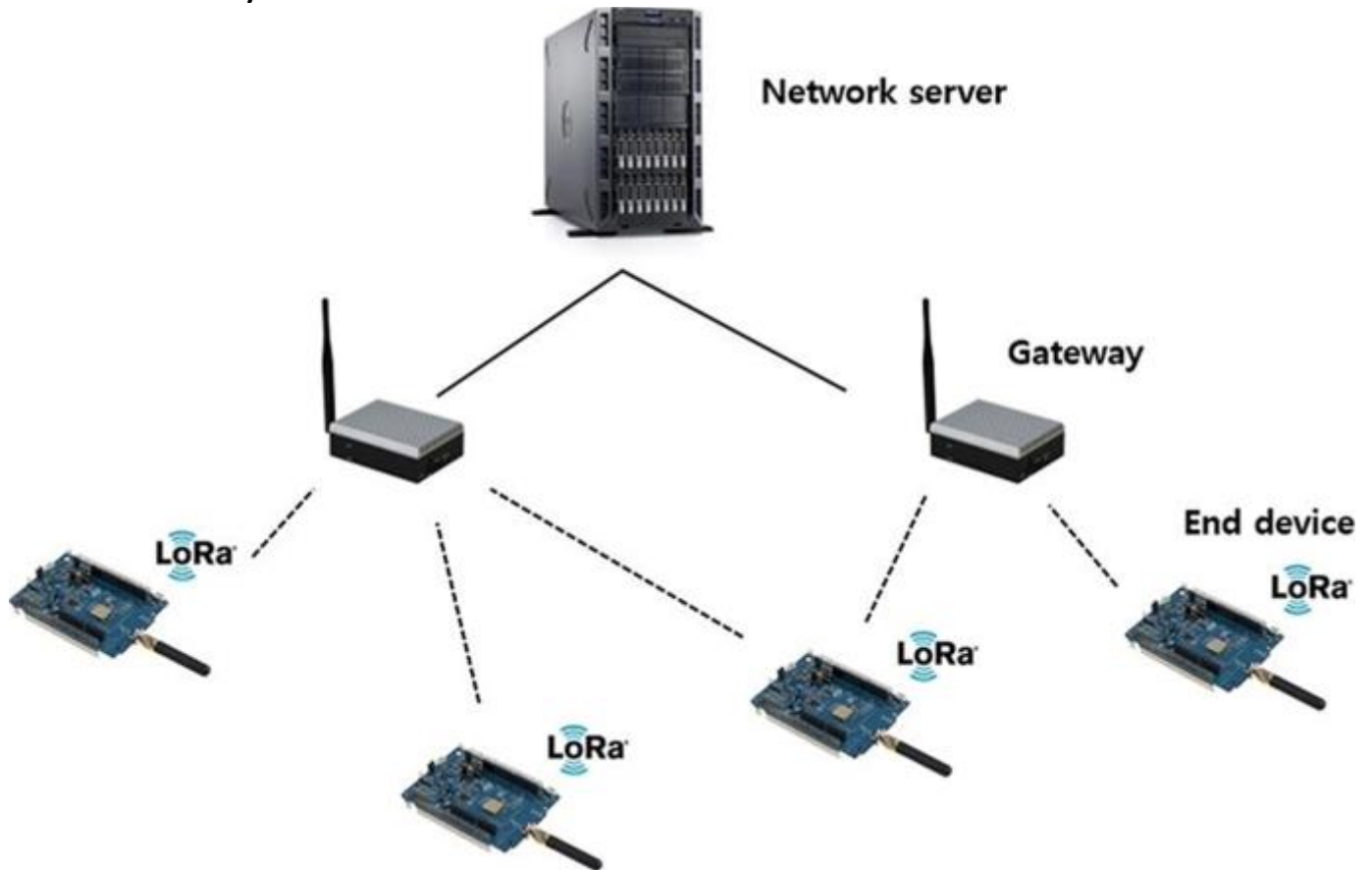


8. LoRa

- LoRa stands for Long range.
- It is a wireless technology designed for long -distance

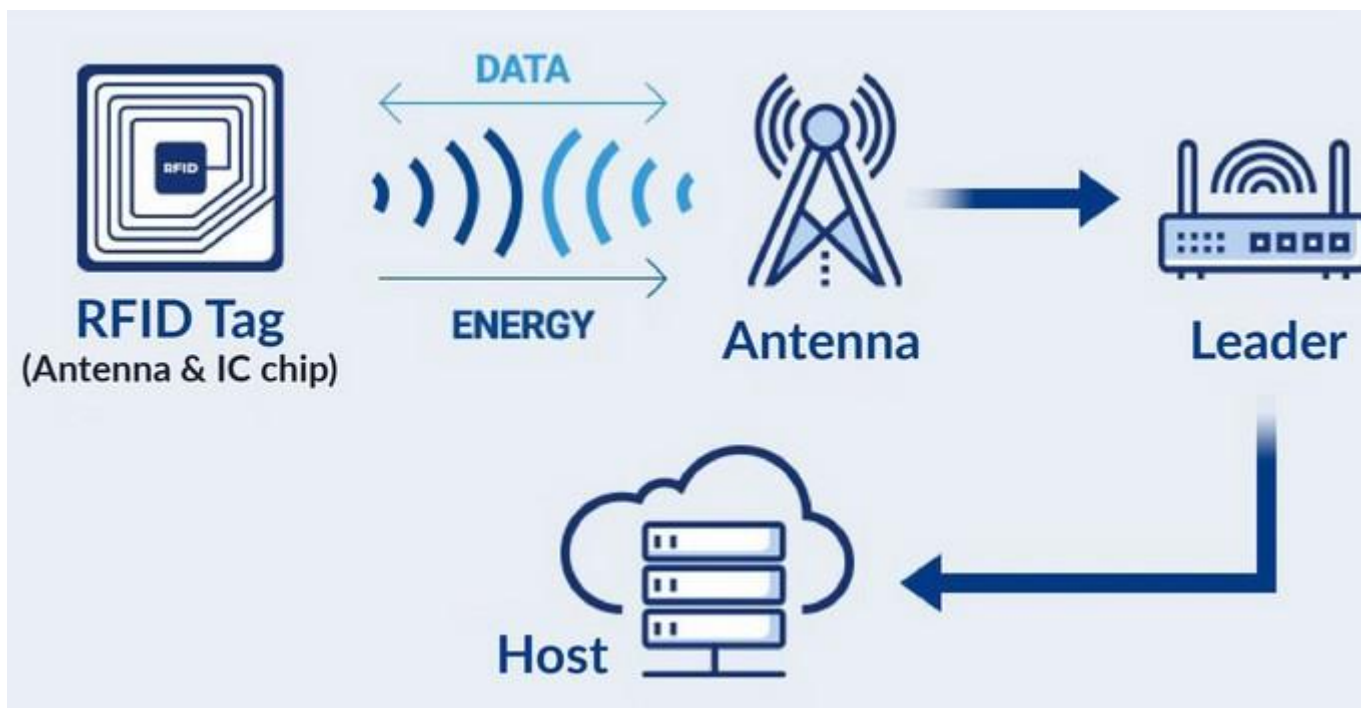
communication.

- It provides low power consumption.
- It is mainly used for IoT devices and sensors.



9. RFID

- RFID stands for Radio frequency Identification.
- It used radio waves to identify and track objects.
- It is used in inventory management, access control, Fastag logistics, and tracking.
- RFID tags can store identification information.
- It range is 100 meter.



10. **Satellite communication**
11. **Infrared(IR)**
12. **Microwave link**
13. **FM/AM radio**
14. **Television broadcasting**
15. **GSM / 2G**

- GSM stands for Global System for Mobile Communications.

- GSM uses SIM cards to identify and authenticate users on the network.
- In this, EDGE (Enhanced Data rates for GSM Evolution) is used to transfer data.
- Facility in mobile communication : Voice calls + SMS + basic data

16. CDMA / 2G & 3G

- CDMA stands for Code Division Multiple Access.
- It is a mobile communication technology that uses unique codes to allow multiple users to share the same frequency.
- It uses EVDO (Evolution Data Optimised) for data transfer.
- Facility in mobile communication : Voice calls + SMS + faster internet + video call + multimedia

17. 4G/LTE

- 4G Stands for fourth Generation of mobile communication.
- It provides high speed mobile internet access.
- LTE (Long term evolution) is the main 4G technology.
- It uses radio waves to communicate b/w mobile devices and base stations.
- Facility in mobile communication : Voice calls + SMS + high-speed internet + video call

18. 5G

- 5G stands for Fifth Generation of mobile communication.
- It provides higher speed and lower latency than 4G.
- It can support a very large number of connected devices.
- It is used for smartphones, IoT, smart cities, and industrial applications.
- Facility in mobile communication : Voice calls + SMS + Very high speed internet + video call + low latency

❖ Generations of Wireless mobile communication

Generation	Introduced	Technology	Data transfer rate
1G	1979, Japan	Analog	Up to 2.4kbps
2G	1990's Finland	Digital (1 st – GSM, 2 nd – CDMA)	50 kbps to 100kbs

3G	2000s, Japan	Digital (CDMA 2000)	384kbps to 14.4 mbps
4G	2010, swedan	LET, Wi-max	100 mbps to 1 gbps
5G	2020 south korea	NR (New radio)	Up to 10Gbps (practical – up to 3 gbps)



Key Points

- ✓ 5G was launched in india by prime minister Narendra modi on 1 October 2022 in Delhi.
- ✓ Wireless phone call made over a cellular network happened in 1973.
- ✓ First mobile call in india was made on 31 july 1995 b/w CM of West bangal Jyoti basu) and Union telecom minister (Sukh ram).

❖ Wireless Positioning/Navigation Technology

19. GPS

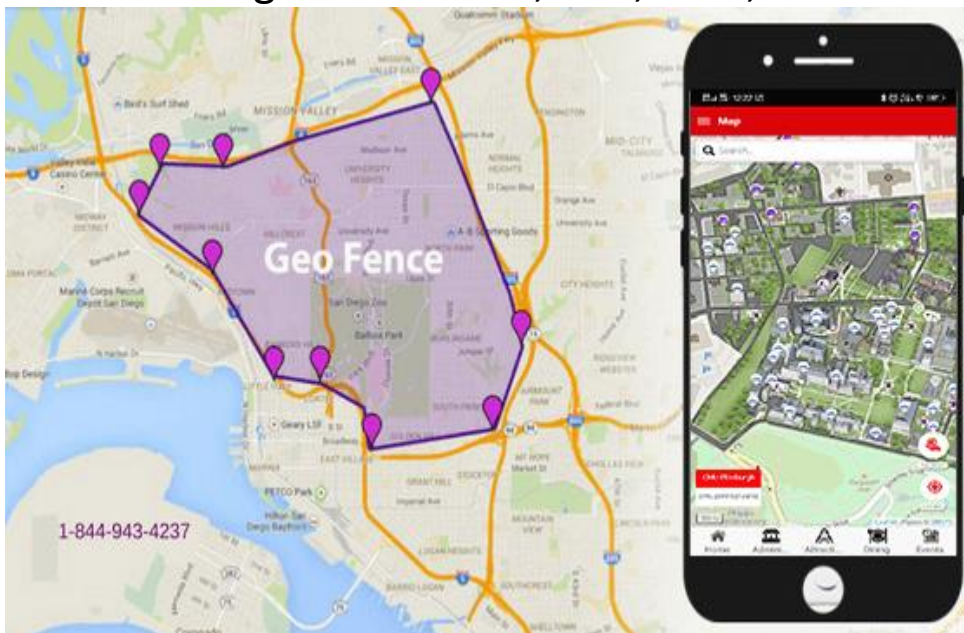
- GPS stands for Global positioning system.
- GPS is a satellite navigation system operated by The USA.
- NavIC is india's regional satellite navigation system developed by

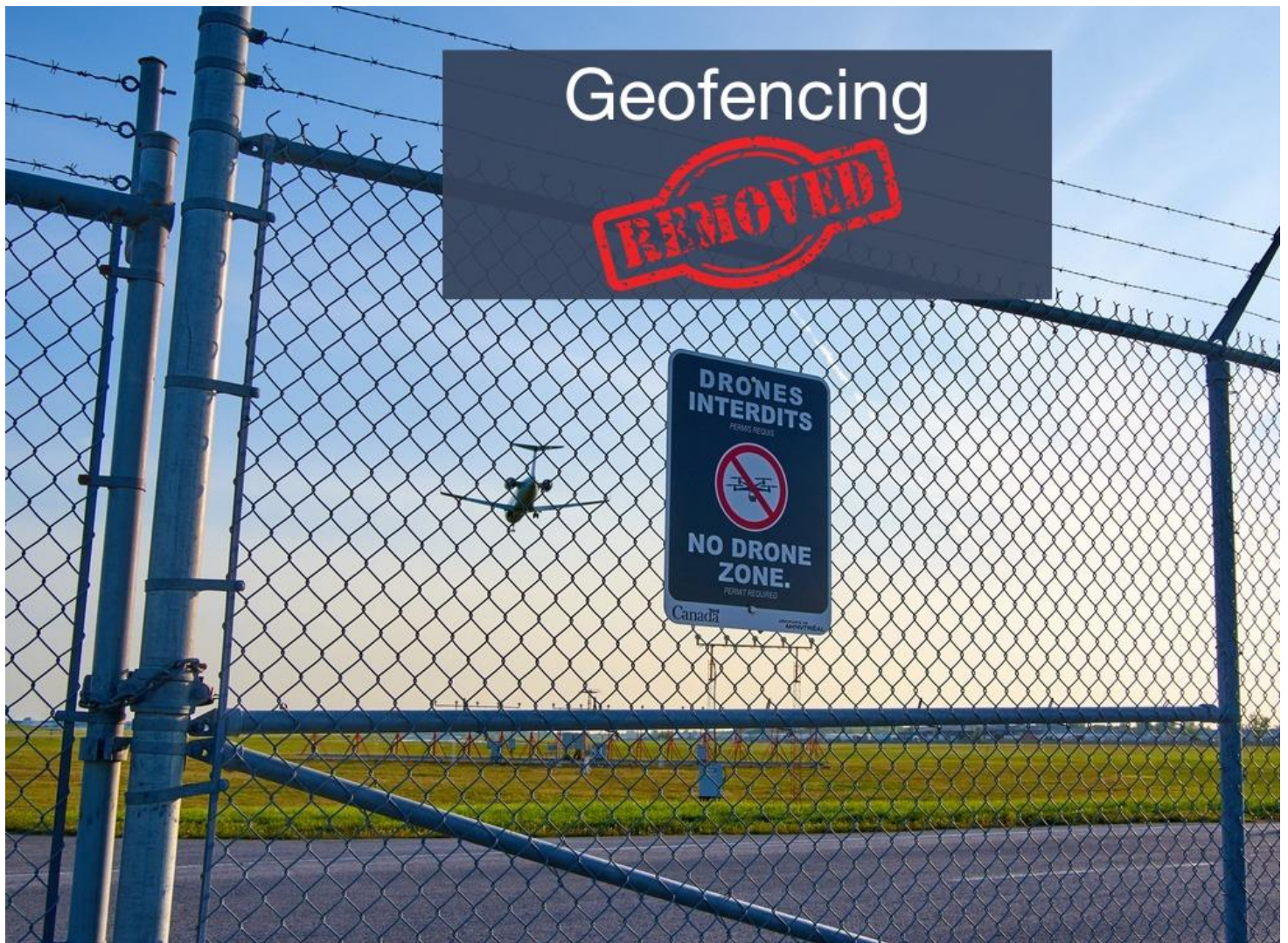
ISRO.

- It uses radio signals from satellites to determine location.
- A GPS receiver receives signals from multiple satellites.
- It can determine latitude, longitude, altitude and time.
- It is used in Google maps, navigation, vehicle tracking and smartphones.
- GPS Operational satellites = 31, range = 20,200 KM
- NavIC Operational satellites = 7

❖ What is Geofencing ?

- A geofencing is a virtual geographical area.
- Location based service(LBS)
- Uses in app/software
- Triggered marketing / government alert action.
- Action by email, advertisement, notification.
- Geofencing uses in : Wifi, GPS, RFID, Bluetooth, drones.





❖ What is an IEEE Standard?

- IEEE standard is a set of technical rules and specifications developed by the IEEE (Institute of Electrical and Electronics Engineers) to ensure that different devices and technologies can communicate and work together properly.
- IEEE 802 family standard mainly introduce/define for networking technologies, communication method/standards.



Some important IEEE 802 family standard and its technology name

1. IEEE 802.1 – LAN / MAN Protocol (Bridging)
2. IEEE 802.2 – Logical Link Control
3. IEEE 802.3 – Ethernet (nic,switch,router,ports) technology
4. IEEE 802.4 – Token Bus
5. IEEE 802.5 – Token Ring
6. IEEE 802.6 - MAN
7. IEEE 802.7 - Broadband
8. IEEE 802.8 – Fiber optics
9. IEEE 802.9 – Integrity voice / Data network
10. IEEE 802.10 – Network security
11. IEEE 802.11 – WLAN (wifi) technology
12. IEEE 802.12 – Demand Priority
13. IEEE 802.13 – Not Used
14. IEEE 802.14 – Cable modems
15. IEEE 802.15 – Wireless PANs
16. IEEE 802.15.1 - Bluetooth
17. IEEE 802.15.4 - Zigbee
18. IEEE 802.15.5 – Mesh Network
19. IEEE 802.16 – WI-Max / WMAN (broadband wireless technology)

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